1) $L_{CE} = -\sum_{i=1..k} y_i \log O_i = -y_1 \log O_1 - y_2 \log O_2$

$\cancel{L_{BCE} = -y_1 \log \hat{y}_1 - (1-y}$

$k=2$

$O_1 + O_2 = 1 \Rightarrow O_2 = 1 - O_1$

$y_1 + y_2 = 1 \Rightarrow y_2 = 1 - y_1$

$L_{CE} = -y_1 \log O_1 - (1-y_1) \log(1-O_1)$

$= -y_1 \log \hat{y}_1 - (1-y_1) \log(1-\hat{y}_1) = L_{BCE}$

2) $\frac{\partial L}{\partial z_i} = \sum_{j=1}^{k} \frac{\partial L}{\partial O_j} \frac{\partial O_j}{\partial z_i}$

$\frac{\partial L}{\partial O_j} = \frac{\partial}{\partial O_j}\left(-\sum_{m=1}^{k} y_m \log O_m\right) = \frac{\partial}{\partial O_j}(-y_j \log O_j) = -\frac{y_j}{O_j}$

$$\frac{\partial O_j}{\partial z_i} = \begin{cases} \dfrac{\exp(z_i)\sum_{m=1}^{k}\exp(z_m) - \exp(z_i)\exp(z_i)}{\left(\sum_{m=1}^{k}\exp(z_m)\right)^2} = O_i(1-O_i) & (j=i) \\ \dfrac{0 - \exp(z_j)\exp(z_i)}{\left(\sum_{m=1}^{k}\exp(z_m)\right)^2} = -O_j O_i & (j \neq i) \end{cases}$$

$\Rightarrow \frac{\partial L}{\partial z_i} = \left(-\frac{y_i}{O_i}\right)[O_i(1-O_i)] + \sum_{j \neq i}\left(-\frac{y_j}{O_j}\right)(-O_j O_i)$

$\cancel{= -y_i - O_i - y_i}$

$= -y_i + O_i\left(y_i + \sum_{j \neq i} y_j\right)$

$= O_i - y_i$